

Statistical & Error Analysis of Website Response Time

A Numerical Analysis Approach to Accuracy, Precision & Error Quantification

Numerical Analysis - Project Work 1 (Ex802-Co4)

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Why Measure Response Time Rigorously?

- Website response time directly affects **user experience**, **SEO ranking**, and **conversion rate**.
- Industry SLAs commonly require server response under a fixed benchmark ($T = 250.0000$ ms).
- Different monitoring tools can report *different* readings for the *same* event due to calibration, sampling jitter, or network variance.
- This project treats response-time monitoring as a **numerical analysis problem**: comparing five measurement series for **accuracy**, **precision**, and **error**.

Core Question

Which engineer's monitoring tool can be trusted as the team standard - and which need recalibration?

Scenario

Five IT engineers independently measure the **response time (in milliseconds)** of the same company website each day, using browser-based monitoring tools.

- **Engineers (instruments):** E1, E2, E3, E4, E5
- **Observations:** 28 readings per engineer (ms, 4-decimal precision)
- **Reference (true) value:** $T = 250.0000$ ms (server-side benchmark)
- **Goal:** Quantify accuracy, precision, and error characteristics of each monitoring tool

Full Dataset - Response Time in ms (Obs. 1–14)

Obs.	E1	E2	E3	E4	E5
1	249.8123	251.2034	245.1023	250.0512	255.3021
2	250.1045	249.6781	244.8912	249.9678	254.8765
3	249.9567	250.8923	245.3201	250.0234	255.6123
4	250.0234	251.0456	244.9876	249.9912	254.9456
5	249.8901	249.5123	245.1789	250.0345	255.2789
6	250.2415	251.9289	245.1482	249.8566	255.0749
7	249.9482	250.5969	245.4210	250.1177	255.0598
8	250.0161	249.6946	245.1315	249.9441	254.9174
9	250.0670	250.3658	245.0154	249.9823	255.0645
10	249.9043	249.1567	245.3719	250.1119	255.6159
11	250.0119	251.2999	245.1269	250.1182	255.2848
12	250.0115	250.3586	245.1070	249.9509	254.9964
13	249.7730	250.1493	245.1291	250.0040	254.9655
14	250.1500	251.3147	245.0728	250.0433	254.8720

All values recorded to 4-decimal precision. $T = 250.0000$ ms is the true reference value. (Continued on next slide.)

Full Dataset - Response Time in ms (Obs. 15–28)

Obs.	E1	E2	E3	E4	E5
15	250.0933	249.5659	245.0469	249.9529	255.1670
16	249.9265	250.9698	244.8822	249.7983	255.1711
17	249.9883	249.3005	245.1655	250.0355	254.7342
18	250.0803	250.2009	245.0783	250.0589	255.1936
19	249.9761	249.8528	245.2667	250.0122	254.9422
20	249.9786	251.5648	245.0382	249.8816	255.4175
21	249.8140	251.7601	244.8132	249.9526	255.1152
22	250.0870	250.4145	245.0776	249.9141	255.4868
23	250.0285	251.1659	245.3408	249.9425	254.9688
24	250.0489	250.5105	245.0360	250.0981	255.0093
25	249.8040	250.9908	244.9619	250.1208	255.1492
26	250.2361	250.1426	244.9175	250.0369	255.3293
27	250.0326	249.5291	244.9385	250.0252	255.2126
28	249.9589	249.4683	245.0480	250.0377	254.7842
$\bar{x}$	249.9987	250.4512	245.0934	250.0023	255.1267

Tools Used

- **Python 3** (NumPy, Pandas)
- **SciPy** - t-distribution, percentiles
- **Matplotlib** - all 7 visualizations

Sample Size

- $n = 28$ observations per engineer
- Degrees of freedom = $n - 1 = 27$
- 4-decimal precision recording

Analysis Pipeline

1. Data Import
2. Descriptive Statistics
3. Visualization (7 charts)
4. Error Metrics
5. Accuracy / Precision Classification
6. Significance Testing
7. Conclusions & Recommendations

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \tilde{x} = \frac{x_{(n/2)} + x_{(n/2+1)}}{2}$$

Engineer	Mean (ms)	Median (ms)
E1	249.9987	250.0117
E2	250.4512	250.3902
E3	245.0934	245.0780
E4	250.0023	250.0178
E5	255.1267	255.0950

Observation

E3 and E5 means are far from $T = 250.0000$ ms - indicating systematic bias (calibration issue), not random noise.

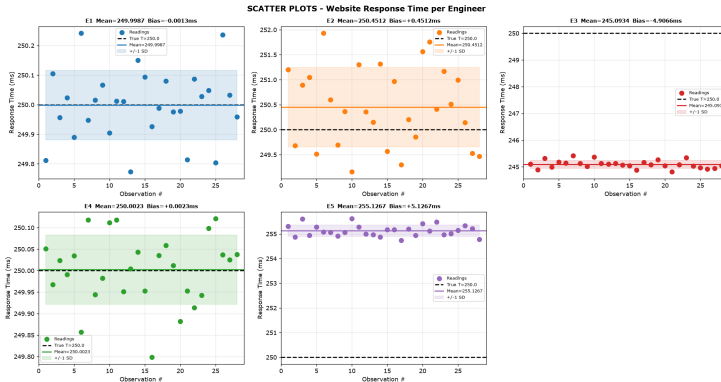
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2} \quad CV = \frac{s}{\bar{x}} \times 100\%$$

Eng.	Variance	Std Dev	Q_1	Q_3	IQR	CV (%)
E1	0.0138	0.1177	249.94	250.07	0.1275	0.0471
E2	0.6329	0.7955	249.69	251.08	1.3852	0.3176
E3	0.0227	0.1506	245.01	245.15	0.1441	0.0614
E4	0.0065	0.0808	249.95	250.05	0.0931	0.0323
E5	0.0529	0.2299	254.96	255.28	0.3199	0.0901

Most Stable Instrument

E4 has the lowest CV ($\approx 0.032\%$) - the most precise and reproducible instrument in the team.

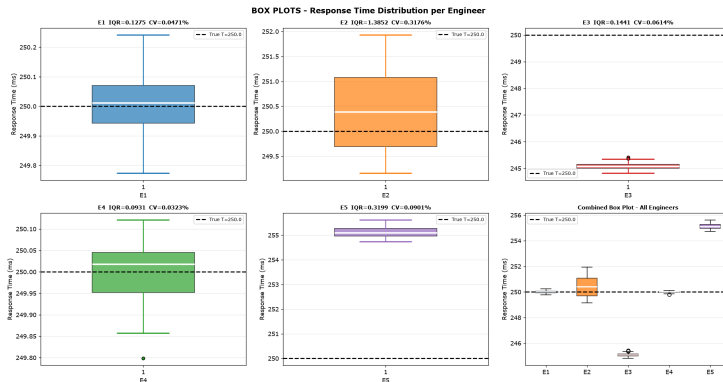
Visualization 1 - Scatter Plot



Reading

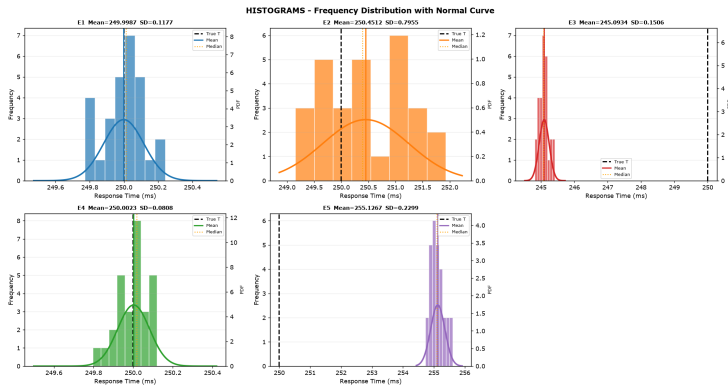
E1 and E4 scatter tightly around $T = 250$; E3 and E5 are entire bands shifted away; E2 shows the widest vertical spread among the on-target group.

Visualization 2 - Box Plot Comparison



Reading

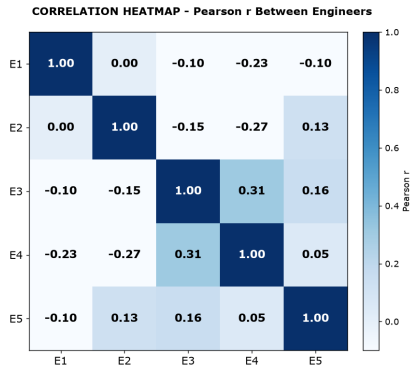
Narrow boxes near the dashed true-value line (E1, E4) indicate good accuracy *and* precision; boxes shifted away (E3, E5) indicate bias; E2's tall box shows the largest spread among the near-target engineers.



Reading

E1, E3, E4 show tight bell-shaped distributions; E3's curve sits entirely left of T ; E5's curve sits entirely right of T ; E2 is the most spread-out distribution.

Visualization 4 - Correlation Heatmap



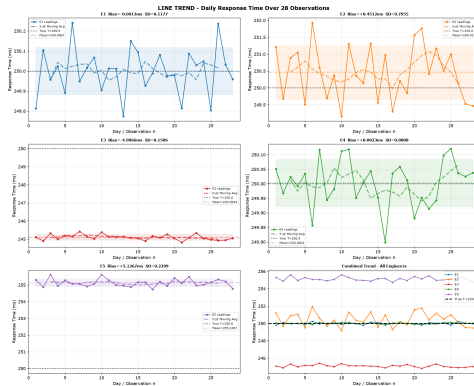
Interpretation

All pairwise correlations are weak ($|r| \leq 0.31$, strongest pair E3–E4).

Day-to-day fluctuations behave like **independent random noise** rather than a shared external factor (e.g. network congestion).

Each engineer's series can be treated as a **statistically independent sample**.

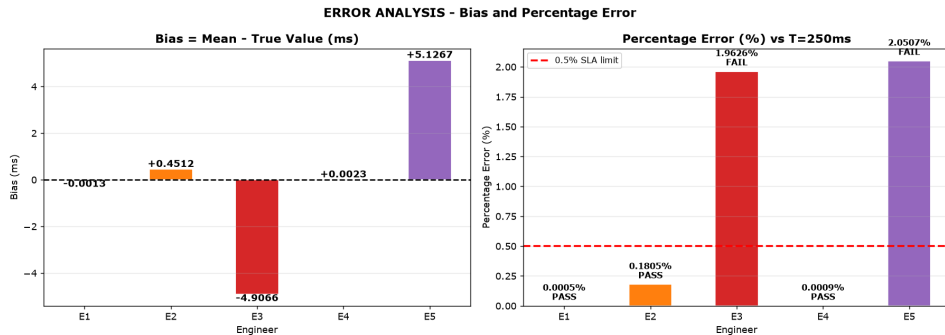
Visualization 5 - Line Trend (28-Day Monitoring)



Reading

5-point moving averages confirm stable means with no long-term drift - each engineer's bias is consistent and systematic, not a worsening trend.

Visualization 6 - Error Analysis (Bias & Percentage Error)



Key Result

Against the 0.5% SLA limit: **E3** and **E5** clearly **FAIL**; E1, E2, and E4 **PASS**.

Absolute, Relative & Percentage Error

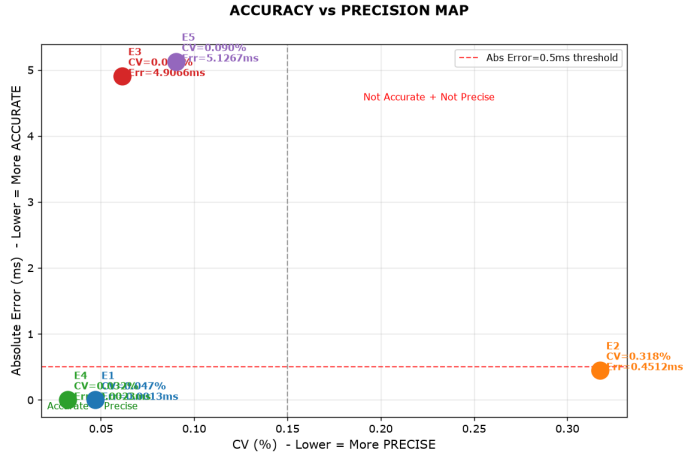
$$E_a = |T - \bar{x}| \quad E_r = \frac{E_a}{|T|} \quad E_{\%} = E_r \times 100$$

Engineer	Mean (ms)	E_a (ms)	$E_{\%}$	Status
E1	249.9987	0.0013	0.0005%	✓ PASS
E2	250.4512	0.4512	0.1805%	✓ PASS
E3	245.0934	4.9066	1.9626%	✗ FAIL
E4	250.0023	0.0023	0.0009%	✓ PASS
E5	255.1267	5.1267	2.0507%	✗ FAIL

Iterative Stopping Criterion

$$\left| \frac{x_{\text{new}} - x_{\text{old}}}{x_{\text{new}}} \right| \times 100 < \varepsilon_s$$

Visualization 7 - Accuracy vs. Precision Map



Accuracy vs. Precision - Summary

Definitions

Accuracy: closeness of $\bar{x}$ to true value T .

Precision: closeness of repeated readings to each other (low spread).

$$\text{Bias} = \bar{x} - T$$

Eng.	Accurate?	Precise?
E1	Yes	Yes
E2	Yes	No
E3	No	Moderate
E4	Yes	Yes (best)
E5	No	Moderate

Key Result

E3 and E5 are biased but reasonably *precise* ($CV = 0.06\%$ and 0.09%); **E2** is paradoxically the *least* precise instrument of all five ($CV = 0.32\%$) despite being accurate enough to pass the SLA - its bias is small, but its day-to-day spread is the largest in the team.

Statistical Significance of Bias - One-Sample t -Test

$$H_0 : \mu = T \quad H_1 : \mu \neq T \quad t = \frac{\bar{x} - T}{s/\sqrt{n}}, \quad df = n - 1 = 27$$

Engineer	t -statistic	$ t $ vs $t_{0.025,27} = 2.052$	Conclusion	
E1	-0.06	Below critical	Not significant	Just luck, tool is fine
E2	+3.00	Above critical	Significant	Real bias, needs check
E3	-172.39	Far above	Significant (extreme)	
E4	+0.15	Below critical	Not significant	
E5	+118.03	Far above	Significant (extreme)	

Important Nuance

E2 *passes* the 0.5% SLA threshold, yet its 0.45 ms bias is **statistically significant** ($t = 3.00 > 2.052$) - consistent and repeatable. A calibration check is recommended.

Round-off Error

From finite decimal representation:

$$e_{\text{round}} = x_{\text{true}} - x_{\text{stored}}$$

At 4 d.p., $e_{\text{round}} \leq 5 \times 10^{-5}$ ms - negligible.

Truncation Error

From approximating infinite processes (e.g. Taylor series):

$$E_t \approx \frac{h^2}{2} f''(\xi)$$

Order of Error $\mathcal{O}(h^n)$

Method	Order	Typical error
Forward diff.	$\mathcal{O}(h)$	5×10^{-3}
Central diff.	$\mathcal{O}(h^2)$	2×10^{-5}

Higher order $\Rightarrow$ faster error reduction as $h \rightarrow 0$.

if h is 0.1 then error is 0.1

if h^2 is 0.1 then error is 0.01

Python Implementation - Statistical Analysis

```
import numpy as np

E1 = [249.8123, 250.1045, 249.9567, 250.0234, 249.8901, 250.2415, 249.9482,
      250.0161, 250.0670, 249.9043, 250.0119, 250.0115, 249.7730, 250.1500,
      250.0933, 249.9265, 249.9883, 250.0803, 249.9761, 249.9786, 249.8140,
      250.0870, 250.0285, 250.0489, 249.8040, 250.2361, 250.0326, 249.9589]
T = 250.0000

arr      = np.array(E1)
mean     = np.mean(arr)
median   = np.median(arr)
std      = np.std(arr, ddof=1)
var      = np.var(arr, ddof=1)
q1,q3    = np.percentile(arr, [25, 75])
cv       = std / mean * 100
abs_err  = abs(T - mean)
pct_err  = abs_err / T * 100
t_stat   = (mean - T) / (std / np.sqrt(len(arr)))

print(f"Mean={mean:.4f}   Median={median:.4f}   Std={std:.4f}")
print(f"IQR={q3-q1:.4f}   CV={cv:.4f}%   Error={pct_err:.4f}%   t={t_stat:.2f}")
```

Program Output Summary

Dataset Loaded: 28×5 observations — True value $T = 250.0000$ ms

	E1	E2	E3	E4	E5
1	249.8123	251.2034	245.1023	250.0512	255.3021
2	250.1045	249.6781	244.8912	249.9678	254.8765
...					

(28 rows x 5 cols)

Statistical Results

Eng	Mean	Bias	PctErr	t-stat	Status
E1	249.9987	-0.0013	0.0005%	-0.06	PASS
E2	250.4512	+0.4512	0.1805%	+3.00	PASS
E3	245.0934	-4.9066	1.9626%	-172.39	FAIL
E4	250.0023	+0.0023	0.0009%	+0.15	PASS <-- BEST
E5	255.1267	+5.1267	2.0507%	+118.03	FAIL

Key Findings

- **E1 and E4** are accurate and precise; their bias is **not** statistically significant.
- **E2** passes the SLA threshold but carries a statistically significant bias ($t = 3.00$) — a real, repeatable offset worth investigating.
- **E3 and E5** show large, highly significant bias ($> 1.9\%$ error) — almost certainly calibration faults.
- Correlation analysis confirms all five series fluctuate **independently**; no shared external noise detected.
- Round-off error is negligible at 4-decimal precision; truncation error shrinks rapidly with higher-order methods.

Lowest absolute error ($E_a = 0.0023$ ms, $E_{\%} = 0.0009\%$) · Lowest CV (0.032%) · Bias not statistically significant ($t = 0.15$).

Adopt E4's measurement protocol as the team standard.

Engineer Performance Scorecard

Eng.	$E_{\%}$	CV (%)	SLA	Accurate?	Precise?	Bias Significant?
E1	0.0005	0.0471	✓ PASS	Yes	Yes	Not significant
E2	0.1805	0.3176	✓ PASS	Yes	No	Significant
E3	1.9626	0.0614	✗ FAIL	No	Moderate	Significant (extreme)
E4	0.0009	0.0323	✓ PASS	Yes	Yes	Not significant
E5	2.0507	0.0901	✗ FAIL	No	Moderate	Significant (extreme)

★ Winner: E4

- **Smallest** percentage error (0.0009%)
- **Smallest** spread / CV (0.0323%)
- **No** statistically significant bias
- Passes SLA with the highest margin

Action Plan

- **Immediate:** Recalibrate E3 & E5
- **Schedule:** Calibration check for E2
- **Adopt:** E4's protocol as standard
- **Use:** Central-difference methods for derived metrics

Thank You
